DEEP LEARNING: THE BASICS

A short, simple introduction to neural networks and how they learn

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1. What is Deep Learning?

Deep learning is a type of artificial intelligence (AI) that teaches computers to find patterns in data. It uses systems inspired by the way connected nerve cells work in the brain. With enough useful examples, a deep-learning model can learn tasks such as recognising images, understanding speech, translating languages, or making predictions. “Deep” usually means the network has several layers that learn different levels of patterns.

2. AI vs Machine Learning vs Deep Learning

Artificial Intelligence (AI): The broad field of making computers perform tasks that normally require human-like intelligence.
Machine Learning (ML): A part of AI where computers learn patterns from examples instead of being programmed with every rule.
Deep Learning (DL): A part of machine learning that uses neural networks with multiple layers to learn complex patterns.

In short: AI is the broad area; machine learning is one way to build AI; deep learning is one kind of machine learning.

3. What is an Artificial Neural Network?

An Artificial Neural Network (ANN) is a computer model made of connected units called neurons. Each connection has a strength, called a weight. The network combines input information, passes it through layers, and produces an output. During training, it adjusts its weights and biases so its predictions become closer to the correct answers.

A basic neural network

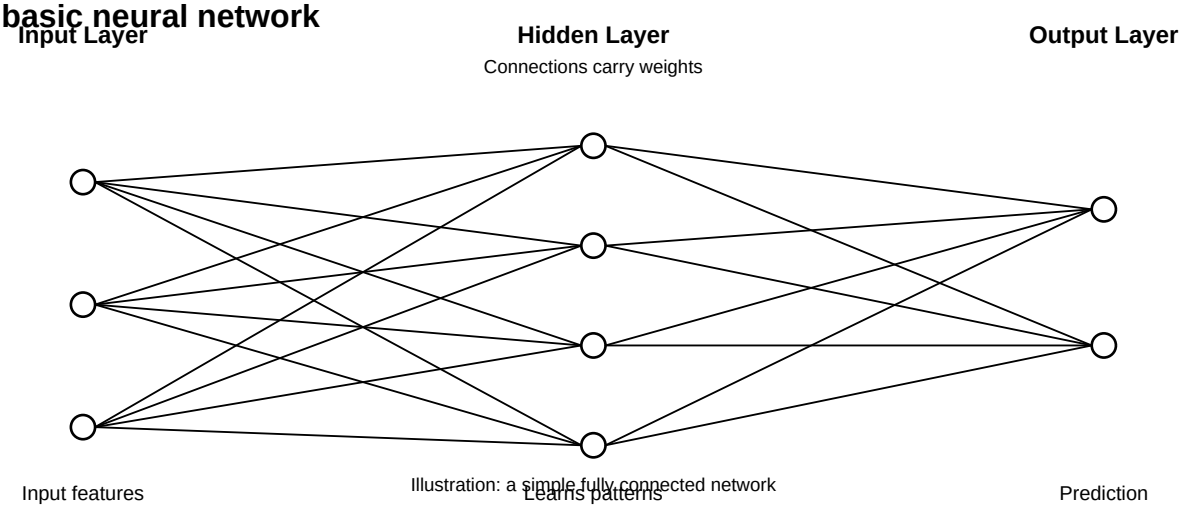

Illustration: a simple fully connected network

The lines show connections between neurons. A real network may have many more neurons and hidden layers.

4. Main parts of a neural network

Neuron: A small processing unit. It receives numbers, combines them, and passes a result onward.
Input layer: Receives the starting information, such as image pixels, a house’s size, or a customer’s age.
Hidden layer: Processes the information and learns useful patterns. A network can have one or many hidden layers.
Output layer: Gives the model’s final result, such as a predicted price, a category, or a probability.

HOW A MODEL LEARNS

The important training terms and the basic learning flow

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5. Important terms

Weights: Numbers that control how strongly one neuron affects another. Training changes these values to improve predictions.
Bias: An extra adjustable number that helps a neuron shift its result, giving the network more flexibility.
Activation function: A rule applied to a neuron's result. It helps the network learn complex, non-straight-line patterns. Common examples include ReLU and sigmoid.
Epoch: One full pass through all the training examples. Models are often trained for many epochs.
Loss: A score that measures how far the model's prediction is from the expected answer. Lower loss usually means a better fit to the training examples.
Optimizer: The method that updates the model's weights and biases to reduce loss. A common optimizer is Adam.

6. Basic working flow

Input → Neural Network → Prediction → Loss → Optimization → Improved Model

1. Input: The model receives training data, such as pictures labelled "cat" or "dog."

2. Neural network: The input passes through the layers. Neurons use weights, biases, and activation functions to process it.

3. Prediction: The output layer produces the model's answer, for example "cat: 70%."

4. Loss: During training, the prediction is compared with the correct label. The loss score shows how wrong the answer was.

5. Optimization: The optimizer uses information from the loss and its gradients to adjust weights and biases.

6. Improved model: The cycle repeats over many examples and epochs. The aim is to make useful predictions on new data, not just memorise training examples.

7. A simple example

Imagine teaching a model to recognise cats. You show it labelled cat and dog pictures. At first, it may guess incorrectly. The loss measures the error, and the optimizer adjusts the network's weights and biases. After seeing many examples, the model may learn patterns such as ears, eyes, and shapes. It then uses those learned patterns to classify a new picture. Its accuracy depends on the quality and variety of the data and on how the model is trained.

Key idea

A neural network learns by making predictions, measuring its errors, and adjusting its internal numbers. This process repeats until the model performs well enough for its intended task. Lower training loss alone does not guarantee good results on new data, so models should also be tested on examples they did not train on.